

Packaging Recommendation System

The Packaging Recommendation System is an AI-driven project developed under the Infosys Springboard Internship. It focuses on recommending sustainable and cost-effective packaging materials by analyzing material properties, environmental impact, and cost parameters.

Milestone 1: Week 1–2

Module 1: Data Collection and Management

Eco-friendly packaging material data was collected with attributes including material type, strength, weight capacity, biodegradability score, CO₂ emission score, and recyclability percentage. Industry product categories such as electronics, food, and cosmetics were incorporated. A PostgreSQL database schema was designed, and CSV/Excel datasets were integrated and validated.

Module 2: Data Cleaning and Feature Engineering

Missing values were handled, and numerical features were normalized. Categorical variables were encoded. Feature engineering was performed to compute CO₂ Impact Index, Cost Efficiency Index, and Material Suitability Score. Data quality was validated using summary statistics.

Milestone 2: Week 3–4

Module 3: Machine Learning Dataset Preparation

The dataset was split into training and testing sets. Relevant ML features such as material safety, strength, and shipping category were selected. Target variables for cost prediction and CO₂ impact prediction were generated. Data pipelines including scaling were prepared.

Module 4: AI Recommendation Model (ML-Based)

Random Forest Regressor was used for cost prediction, and XGBoost Regressor was used for CO₂ footprint prediction. Models were evaluated using RMSE, MAE, and R² score. A final material ranking and recommendation system was developed based on model predictions.

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